

Dimension groups in D365 F&O

The screenshot shows the D365 F&O interface. At the top, the navigation bar includes 'Finance and Operations', 'Product information management', 'Products', and 'Released products'. Below this, a toolbar contains icons for 'Edit', 'New', 'Delete', 'Proposed change', and 'Workflow'. A secondary toolbar has tabs for 'Product', 'Purchase', 'Sell', and 'Manage inv'. The main area is divided into sections: 'Maintain' (Apply template, Validate), 'New' (Product Template), 'Product master' (Released product variants, Product dimensions), and 'Languages' (Translations). A 'Dimension groups' dropdown menu is open, showing a list of dimension groups: 'Product dimension group', 'Storage dimension group' (with 'WMS' selected), and 'Tracking dimension group' (with 'None' selected). An 'OK' button is visible at the bottom right of the dialog.

The inventory dimensions can be assigned to inventory dimension groups:

- Product dimension group:
 - **Configuration**
 - **Color**
 - **Size**
- Storage dimension group:
 - **Site**
 - **Warehouse**
 - **Location**
 - **Pallet ID**
- Tracking dimension group:
 - **Batch number**
 - **Serial number**

Storage dimension groups and tracking dimension groups are set up in a similar manner. However, product dimension groups are configured and used differently.

Dimension groups in D365 F&O

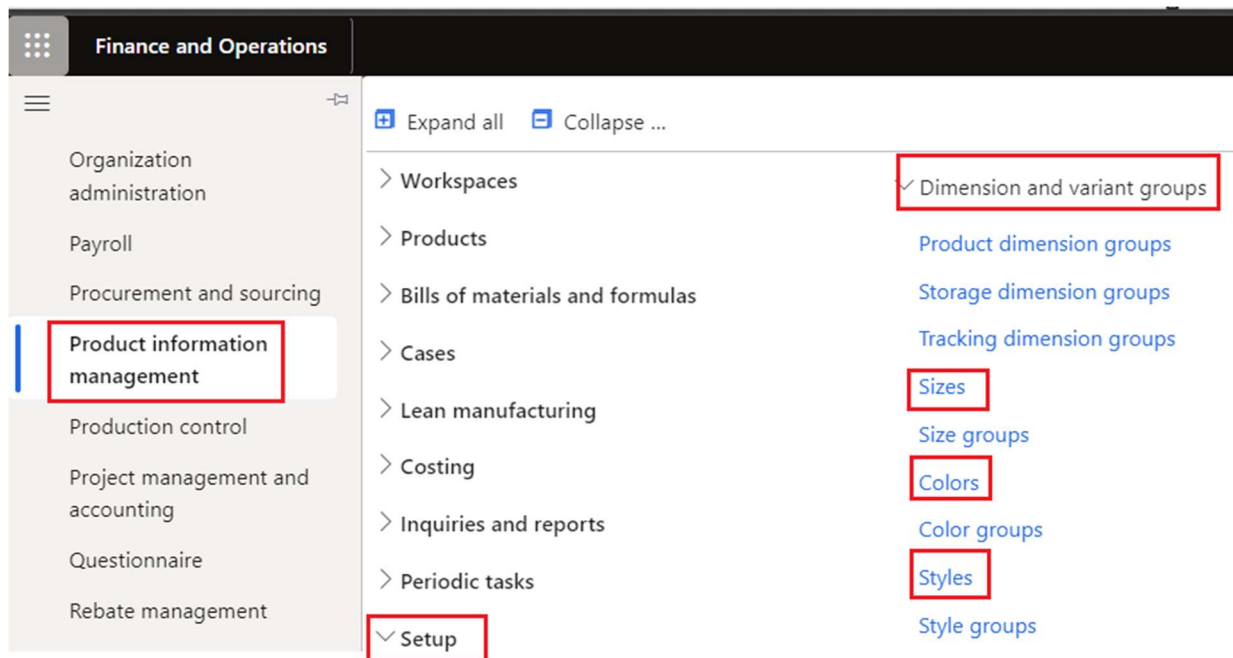
1)Product dimensions Groups

There are five product dimensions: color, configuration, size, style, and version. You combine product dimensions in dimension groups and assign dimension groups to product masters. The combinations of product dimensions determine how product variants are defined.

Product dimensions are characteristics that serve to identify a product variant. You can use combinations of product dimensions to define product variants. You must define at least one product dimension for a product master to create a product variant.

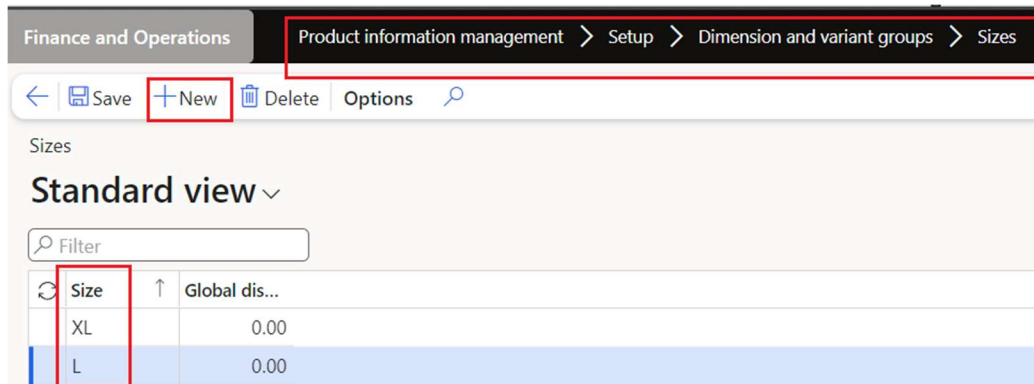
A product variant can be generated based on the product dimension values.

Product dimension values for the size, color, and style dimensions can be created in the following locations:

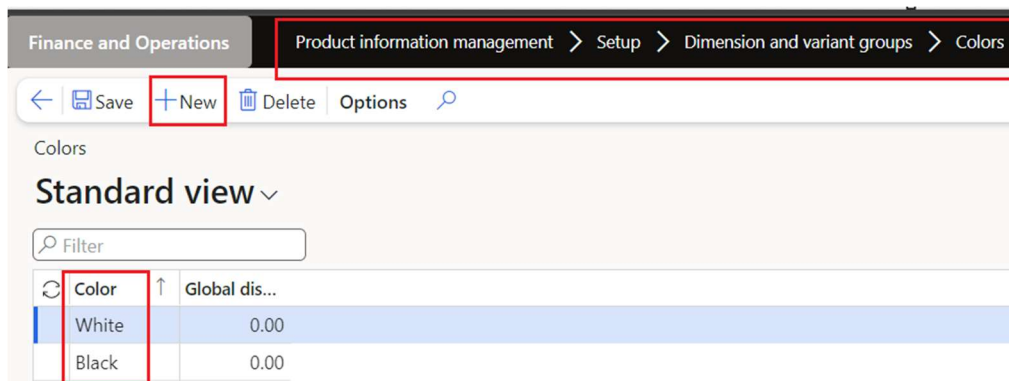


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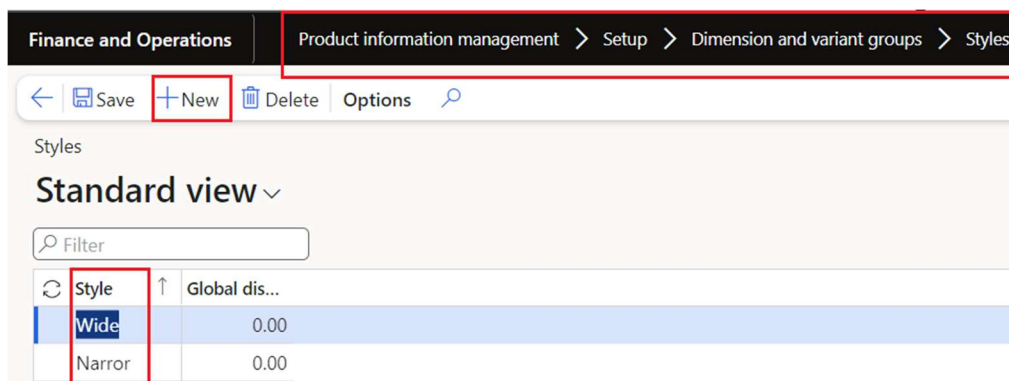
- Size page (Product information management > Setup > Dimension and variant groups > Sizes)



- Color page (Product information management > Setup > Dimension and variant groups > Colors)



- Style page (Product information management > Setup > Dimension and variant groups > Styles)



Dimension groups in D365 F&O

Navigation: Product information management > Setup > Dimension and variant groups > Product dimension groups

Finance and Operations

Product information management > Setup > Dimension and variant groups > Product dimension groups

Save + New Delete Options

Filter

Retail Kit Configuration

Standard view

Product dimension groups

Name: Retail Kit Description: Configuration Use number nomenclature: No Product variant number nomenclature:

Product dimensions

Name	Active	For purchase prices	For sales prices	Display... ↑
Configura...	☒	☒	☒	0.00
Size	☐	☒	☒	0.00
Color	☐	☒	☒	0.00
Style	☐	☒	☒	0.00

For purchase prices

☒

When this parameter is checked in then it creates different purchase prices in purchase order based on site and warehouses. We can set up prices for items in trade agreement journal. Path **procurement and sourcing > Prices and discounts > Trade agreement journals**

Likewise, if this parameter is blank \ unchecked then it will not be able to setup different prices based on site and warehouses in purchase order.

For sales prices

☒

Select this option to include the dimension as a criterion for determining the sales price for an item. In the Journal lines, price/discount agreement page, you can create a trade agreement where you can associate a sales price with a dimension setup for an item. A dimension must be active to be available in a trade agreement setup. Select the Active option in this page to enable a dimension.

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Example

Changes in PO lines while parameter is checked(✓) based on site and warehouse:

We can see two different prices for same item for two different warehouses. These changes happened when ticked the parameter "For Purchase Prices" and posted it in trade agreement journal.

Purchase order lines									
+ Add line + Add lines Add products Remove Purchase order line Financials Inventory Product and supply Update line Warehouse FOC Update									
	Line...	Item number	Quantity	Unit	Product name	Sale base price	Unit price	Net amount	Warehouse
✓	1		1.00	PCS	WATCH RING CHANGEABLE LJ-2177C	33.60	30.00	30.00	292
	2		1.00	PCS	WATCH RING CHANGEABLE LJ-2177C	33.60	32.00	32.00	291

There is no changes in PO lines while the parameter is not checked \ blank

Purchase order lines									
+ Add line + Add lines Add products Remove Purchase order line Financials Inventory Product and supply Update line Warehouse FOC Update									
	Line...	Item number	Quantity	Unit	Product name	Sale base price	Unit price	Net amount	Warehouse
	1		1.00	PCS	WATCH RING CHANGEABLE LJ-2177C	33.60	35.00	35.00	292
✓			1.00	PCS	WATCH RING CHANGEABLE LJ-2177C	33.60	35.00	35.00	291

2. Storage Dimension groups

storage dimensions represent how you store or retrieve items in your inventory. Obviously, this does allow for very detailed management of your inventory. However, there is indeed a challenge in the details of these dimensions, specifically when you have items such as food or chemicals, which require special handling.

Now, let's break down the parameters of storage dimensions.

1. **Site:** Refers to your facilities. Either for storage, manufacturing, or sales.
2. **Warehouse:** Determined by your racking or shelving system, hence, how you store your items in the warehouse.
3. **Location:** For example, if you dealt with chemicals, you would not want to store chemicals closely if they had adverse reactions to each other.

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4. **Inventory status:** Are you storing raw materials? Do you have a production line? Is your finished product ready for sale? Whenever an item arrives or leaves your warehouse, this is where the business should represent it.
5. **License plate:** There is an exciting feature of Dynamics 365, **license plate location positioning**. If a business used pallet racking, there would be a license plate in each pallet. Therefore, using this feature adds a sequence number to each license plate within that location. Furthermore, as a warehouse manager, this would be useful for several picking methods, as you would know beforehand where and how (whether the pallets are front-facing or not) each pallet is.

How do you use Storage Dimensions?

Understandably, tracking dimensions could seem to be more useful. Nevertheless, there is an amazing potential for warehouse management optimization by using storage dimensions.

- **Primary Stocking**

You can set up a **primary stocking** dimension. If you do so, it will appear automatically on the **On-hand** and **Reservation** pages. Therefore, it serves the purpose of ordering items and re-reserving them based on the physical item receipts. You must specify the storage dimension group used to update item receipts or issues. Moreover, since the basis for item reservation is the dimension, if you used the Warehouse dimension, the automatic reservation would only reserve items from that chosen Warehouse.

Therefore, if you received a sales order line using Warehouse: B as primary stocking and Location: 10 as secondary stocking, the automatic reservation would only reserve items at this Warehouse. Whereas Location: 10 would serve as the secondary stocking dimension. However, keep in mind that the reservations would not apply re-reserve automatically after receiving the item if you changed the warehouse when updating the physical receipt.

- **Physical Inventory**

By selecting the check box with this parameter, you would specify its particular dimension entry via **on-hand inventory**. Therefore, the system would check this field to determine whether the item is available in the inventory or not.

- **Financial Inventory**

This parameter serves two purposes when you are not operating with a financially negative inventory. Firstly, the system checks item issues with the same logic of the physical inventory parameter, allowing you to determine the item's availability. Secondly, it allows you to determine item consumption when updating invoices, closing inventory, and accounting for the cost value.

- **Coverage plan by Dimension**

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If you select the Coverage plan by dimension on the Storage dimension groups, it will complete the coverage plan for individual items instead of groups.

- **Search items by “Purchase price” or “Sales price.”**

For both checkboxes present on the Storage dimension groups; will search for either purchase price or sales price, based on these rules:

- Search price for each item, product dimensions, and storage dimensions
- Storage dimensions are blank—Search price for each item and product dimensions
- Both product and storage dimensions are blank—Search price for each item

Summary:

When it comes to Storage Dimensions, these are the dimensions:

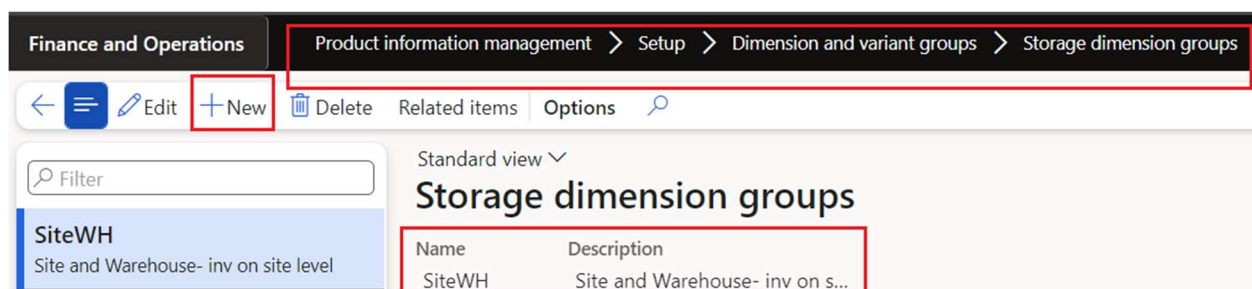
1. Site
2. Location
3. Warehouse
4. Inventory status
5. License plate

And these are its main functions:

1. Primary stocking
2. Physical inventory
3. Financial inventory
4. Coverage plan by Dimension
5. Search items by “Purchase price” or “Sales price.”

Setup Storage dimension group

Navigation: Product information management > Setup > Dimension and variant group > storage dimension groups



Dimension groups in D365 F&O

Storage dimensions		Blank receipt allowed	Blank issue allowed	Physical inventory	Financial inventory	Coverage plan by dimensi...	For purchase prices	For sales price
Name	Active							
Site	✓			✓	✓	✓		
Warehou...	✓					✓		
Location								
Inventory...								
License p...								

Blank receipt allowed

Select this option to indicate that the dimension does not have to be specified when physical receipts are updated. This functionality is useful if, for example, you do not want to specify the serial or lot number upon input, but you do want to do so for picking and output. This option applies only if the Primary stocking option is cleared.

Blank issue allowed

Select this option to indicate that the dimension does not have to be specified when physical issues are updated. This option applies only if the Primary stocking option is cleared.

Physical inventory

A selected option indicates that items that are assigned to this dimension group are tracked physically at this dimension level. Transactions that are generated to track physical transfers are also included in cost. A non-financial transfer is a transfer between two inventory dimensions that is not financially tracked. If items are tracked physically, a limitation for negative inventory must be observed. By using the Physical negative inventory option in the Item model groups page, you can enable negative inventory for items that are associated with the item model group. If negative inventory is not enabled, and an item with this dimension is tracked physically, the item must be available physically to process transactions on the item.

Financial inventory

A selected option indicates that items that are assigned to this dimension group are included in cost calculations at this dimension level.

Coverage plan by dimension

Select this option to include the dimension when you create a coverage plan for a product. To create a coverage plan for a product, select the product in the Released product details page, and then click Item coverage on the Plan tab.

For purchase prices

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Select this option to include the dimension as a criterion for determining the purchase price for an item. In the Journal lines, price/discount agreement page, you can create a trade agreement where you associate a purchase price with a specific dimension setup for an item. A dimension must be active to be available in a trade agreement setup. Select the Active option in this page to enable a dimension.

For sales prices

Select this option to include the dimension as a criterion for determining the sales price for an item. In the Journal lines, price/discount agreement page, you can create a trade agreement where you can associate a sales price with a dimension setup for an item. A dimension must be active to be available in a trade agreement setup. Select the Active option in this page to enable a dimension.

Note - when only site is financial inventory enabled then you can be left with residual financial amounts depending upon the Inventory model group also.

Site specific setup

Show the cost price per site in th...

☒ No

Warehouse specific setup

Use warehouse management pro...

☒ No

WAREHOUSE SPECIFIC SETUP

Mandatory

☒ No

Primary stocking

☒ No

Use warehouse management processes

When using Advance warehouse

Mandatory

Select this option to define the Warehouse dimension as a mandatory dimension for items that are associated with the dimension group. The Site dimension is always mandatory. However, the Location cannot be set to mandatory.

Primary stocking

Dimension groups in D365 F&O

Select this option to define the Warehouse dimension as the primary dimension for items that are associated with the dimension group. If the Warehouse dimension is specified as a primary dimension, this dimension is always locked when items are reserved. It cannot be unlocked.

3. Tracking Dimension groups

Think of tracking dimensions as mechanisms to **ensure tracking, [quality assurance](#), and warranties**. Therefore, these mechanisms come in the form of serial, batch numbers, and ownership. Tracing will allow you to keep track of your products, essentially becoming a handy tool for quality assurance and dealing with [warranties](#). For example, when it comes to certain items that present an inherent risk, such as chemicals, you can prevent losses and item recalls by tracing the items back to the production line, preventing defects. Moreover, think of a customer that had a complaint regarding a certain item. Nevertheless, unless you had accurate control of the dates of manufacturing and sale, you would not know how to deal with that situation.

Furthermore, when it comes to Dynamics 365, you will assign serial and batch numbers to an item. However, these are not mutually exclusive; you can assign both, either or none of them to an inventory item. Moreover, assigning a tracking dimension to an inventory item will be necessary for each product in your system, as it will be the mechanism allowing you to track the item throughout the manufacturing and sales process.

1. Serial Numbers:

The first difference between serial and batch numbers is that batch numbers apply to group items. However, serial numbers apply on a single-item basis.

2. Batch Numbers:

For batch numbers, not only do these apply to groups of items. Therefore, think of the following scenario. If you had a customer to which you sold a batch of products and later found out that these products had an issue, the batch number would issue a **product recall** to your customer. Moreover, you would be able to do so accurately; finding the specific products could have them back with a **vendor return order**.

3. Ownership:

Let's think in practical terms for this one as well. Think of a **consignment** process. Consigned inventory consists of items owned by a third party (a vendor), although stored in your warehouse. Therefore, ownership is essential within dynamics. However, as the ownership changes, it becomes the business' liability to settle vendor invoices and provide adequate payments.

How do you use Tracking Dimensions?

Dimension groups in D365 F&O

Do you remember that we used the term traceability? That is the keyword to understanding the function of tracking dimensions. As such, the trace inventory dimension is the most important function. To use this function, we need to understand the **Lot ID** parameter. Therefore, think of these as transaction numbers and not as an item dimension. Moreover, in essence, a **Lot ID** is an automatically assigned number to each sale or purchase line that you create. However, the **Lot ID** is a requirement to make transactions, unlike optional batch numbers. Furthermore, the **Lot ID** number will correspond with the number sequence in **Lot No.** within the inventory parameters.

- **Trace Inventory Dimensions Function**

The trace inventory dimensions function will allow you to trace inventory items through ownership, serial, and batch numbers and other parameters such as **item number, record ID, Lot ID, and dimension number**. By specifying these parameters in the trace inventory dimensions, you will receive a report. Moreover, this report will display results in a tree graphic containing transactions, associated quality orders, production orders, and batch numbers. Therefore, using this function, you will be able to trace items back to their raw materials all the way until the production and sale of the finished product.

Lastly, amongst its benefits, the tracing function will provide you precise data on customer impact, sales orders, shipping, and information on your production line for each item.

Summary:

When it comes to Tracking Dimensions, these are the dimensions:

1. Serial Numbers
2. Batch Numbers
3. Ownership

And this is its main function:

1. Trace Inventory Dimensions Function

Setup tracking dimension groups

Navigation: Product information management > Setup > Dimension and variant group > Tracking dimension groups

Dimension groups in D365 F&O

Finance and Operations

Product information management > Setup > Dimension and variant groups > Tracking dimension groups

Edit + New Delete Related items Options

Filter

Standard view

Tracking dimension groups

Name	Description
Batch	Batch Enabled

Tracking dimensions

Name	Active	Active in s...	Primary st...	Blank recei...	Blank issue...	Physical in...	Financial i...	Coverage ...	For purcha...	For sales p...	Transfer
Batch numl	✓					✓	✓				
Serial nu...											
Owner						✓	✓				✓

Primary stocking

Select this option to define the Warehouse dimension as the primary dimension for items that are associated with the dimension group. If the Warehouse dimension is the primary dimension, it is always locked when items are reserved. It cannot be unlocked.

Blank receipt allowed

Select this option to indicate that the dimension is not specified when physical receipts are updated. This option is useful if, for example, you want to specify the serial number or lot number for picking and output, but not for input. This option applies only if the Primary stocking option is cleared.

Blank issue allowed

Select this option to indicate that the dimension is not specified when physical issues are updated. This option applies only if the Primary stocking option is cleared.

Physical inventory

A selected option indicates that items that are assigned to this dimension group are physically tracked at this dimension level. Transactions that are generated to track physical transfers are also included in cost calculations. A transfer that occurs between two inventory dimensions, and that is not financially tracked, is called a nonfinancial transfer. If items are physically tracked, there is a limitation on negative inventory. Select the Physical negative inventory option in the Item model groups page to enable negative inventory for items that are associated with the item model group. If negative inventory is not enabled, and an item with this dimension is physically tracked, transactions can only be processed for the item if the item is physically available.

Financial inventory

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A selected option indicates that items that are assigned to this dimension group are included in cost calculations at this dimension level.

Coverage plan by dimension

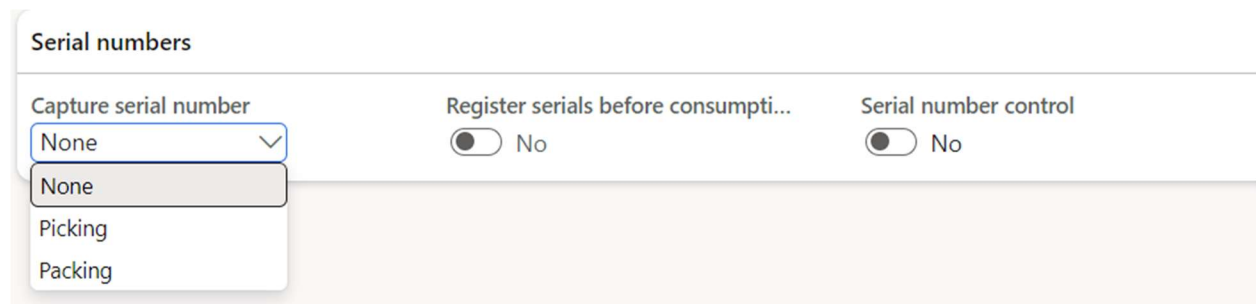
Select this option to include the dimension when you create a coverage plan for a product. To create a coverage plan for a product, select the product in the Released product details page, and then click Item coverage on the Plan tab.

For purchase prices

Select this option to include the dimension as a criterion for the purchase price of an item. In the Journal lines, price/discount agreement page, you can create a trade agreement in which you associate a purchase price with the setup of a dimension for an item. Only active dimensions are available in the setup of a trade agreement. Select the Active option in this page to enable a dimension.

For sales prices

Select this option to include the dimension as a criterion for the sales price of an item. In the Journal lines, price/discount agreement page, you can create a trade agreement in which you associate a sales price with the setup of a dimension for an item. Only active dimensions are available in the setup of a trade agreement. Select the Active option in this page to enable a dimension.



Serial numbers

Capture serial number	Register serials before consumpti...	Serial number control
None	☐ No	☐ No
None		
Picking		
Packing		

Capture serial number

When the field is set to Packing or Picking, the fields Blank receipt allowed and Blank issue allowed are automatically selected and disabled for editing. If the field is set to Picking, the serial number must be captured during the picking process with the mobile device, even though blank receipt and issue are allowed. If the field is set to Packing, the serial number must be

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captured in the manual packing station process. If the field is set to None, then you are not prompted to capture the serial number in the picking or packing process.

Register serials before consumption

Tracks which component serial numbers are used in production orders. When the field is selected, you can postpone the registration of a component serial number until the time of production consumption. This can be useful for tracking non-compliance issues. The field works together with the Capture serial field. For example, when you select Picking in the Capture serial field, the serial number needs to be provided at the picking time. Notice that you should clear the Active check box in the grid of the dimensions in the sales process.

Serial number control

Select this option to indicate that inventory movements must have a quantity of 1. This constraint prevents one item from having two or more identical serial numbers. When items are physically received, or when they are updated as arrived, the physical on-hand quantity and the arrival quantity cannot exceed 1. Based on the value of this field, the Physical inventory option is automatically selected or cleared to handle the control correctly.

Note – Both serial number and Batch number can be manual or Automatic (Number sequence)